

Haobo (Louie) Zhang

🏠 San Jose, CA, US | 📞 206-609-8522 | ✉️ hbxzhang@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 📄 Profolio
Eligible to work under **OPT (F-1)** or **TN** visa—no sponsorship required

WORK EXPERIENCE

Software Engineer Intern

SK Hynix Memory Solutions America

San Jose, CA
Oct 2025 - Present

- Supported enterprise **PCIe SSD** validation and debug, partnering with Application Engineering and system engineers to reproduce failures, isolate root causes, and improve product reliability
- Developed and maintained Python-based test automation to streamline validation workflows, including test orchestration, log collection, and result reporting for system-level testing
- Built AI-assisted engineering utilities using **Retrieval-Augmented Generation (RAG)** and related workflows to accelerate troubleshooting, knowledge retrieval, and validation triage
- Applied ML techniques for log mining and anomaly detection to surface failure signatures, reduce manual log review, and improve debug efficiency

Software Development Engineer Intern

Healthy Claws

New York, NY
Apr 2024 - Aug 2024, June 2025 - Oct 2025

- Developed an enterprise order management platform with **Spring Cloud**, implementing service discovery and inter service communication, and building persistence workflows with Spring Data JPA
- Optimized order query performance by analyzing **MySQL** slow logs with pt-query-digest and tuning composite and covering indexes, reducing latency from 2.5s to 112ms
- Implemented RBAC permission management with **Spring Security** and **JWT**, enabling dynamic role and permission validation and secure access control
- Built standardized logging with Logback and centralized observability via ELK, plus CI/CD with **Docker**, **Jenkins**, and **AWS** to improve release stability and deployment efficiency

Research Engineer Intern

Changjia Fengxing Intelligent Technology Co., Ltd

Jan 2024 - Aug 2024

- Developed a robust **vehicle-infrastructure cooperative** 3D object detection model to address communication latency and sensor synchrony, achieving ~2.5% improvement at 500ms latency compared to the baseline (FFNet) in the AIR-V2X-C dataset, as reported in a 2023 NeurIPS paper
- Developed a **multi-modal collaborative perception system** for autonomous driving by integrating camera and LiDAR data, achieving an 8% performance improvement compared to the original vision-based system
- Collaborated with scientists and domain experts to shape future research directions and experimental focus

Management Information System Analyst

Edith-Min International Inc.

Oakville, ON, Canada
Nov 2020 - Jul 2022

- Defined logical and physical service architecture for a large information system; produced a host and configuration matrix with SLOs, capacity plan, rollout timelines, and cost targets
- Partnered with vendors to design operational databases and established **SAP EDI** integrations with Amazon, Best Buy and Wayfair; added idempotent processing, retries and monitoring
- Built data pipelines for demand **clustering and forecasting** on **200k+** orders using K-Means and ARIMA; generated warehouse placement and inventory recommendations

PROJECTS

JIRAG

Oct 2025 - Present

- Designed a three layer HiRAG knowledge graph in Neo4j and shipped a multi mode query CLI for hierarchical retrieval and QA, returning summaries, expert matches, and historical fixes
- Built a scalable ingestion and infra stack to parse 8,384+ JIRA issues into typed nodes and relations with schema and indexing, and enabled pluggable LLM embedding backends plus Dockerized Neo4j (APOC, GDS)
- Added evaluation harness with golden queries and regression tracking to quantify recall/precision across summary, expert, and fix-history modes
- Implemented caching and embedding reuse strategy, lowering token/compute cost while keeping CLI response times stable

Scalable Distributed Ski Data System

May 2024 - Aug 2024

- Designed and built a distributed cloud system with **Java** and **AWS EC2**, optimizing **JVM** memory management to eliminate memory bottlenecks, enabling the system to scale from 1 million to 5 million ski lift data records
- Developed a client-server architecture using **HTTP API** with a stateless design, eliminating lock contention in a multi-threaded environment
- Integrated **AWS Load Balancer** and **Auto Scaling**, ensuring the system could automatically scale during traffic surges, resolving resource overload issues, and maintaining response times below 100 milliseconds for each request
- Optimized data caching strategy using **Redis**, combining **LRU** strategy and expiration mechanisms to significantly reduce database access frequency, lowering query latency from 200 milliseconds to 50 milliseconds and increasing cache hit rate
- Conducted high-concurrency stress testing with **JMeter**, simulating 4,000 requests per second, optimized thread pool and message queue configurations, resolved deadlock and blocking issues under high load during peak traffic

SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, HTML5/CSS3

Frameworks & Libraries : React, React Native, Next, Node.js, SpringCloud, SpringBoot, TeraTerm, Flask, Django

Databases & Cloud: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, AWS S3, AWS Lambda, Amazon SES, Google Cloud Platform

Other Tools: Nginx, Kafka, RabbitMQ, Docker, Jira, JUnit, CI/CD, Git, Postman, Linux, Conda

EDUCATION

Northeastern University

Master of Science in Computer Science 3.9/4

Seattle, WA
Sept 2022 - May 2025

Brock University

Master of Professional Accounting

St. Catharines, ON, Canada
Aug 2019 - Oct 2020

Roanoke College

Bachelor of Business Administration

Salem, VA
Aug 2015 - May 2019